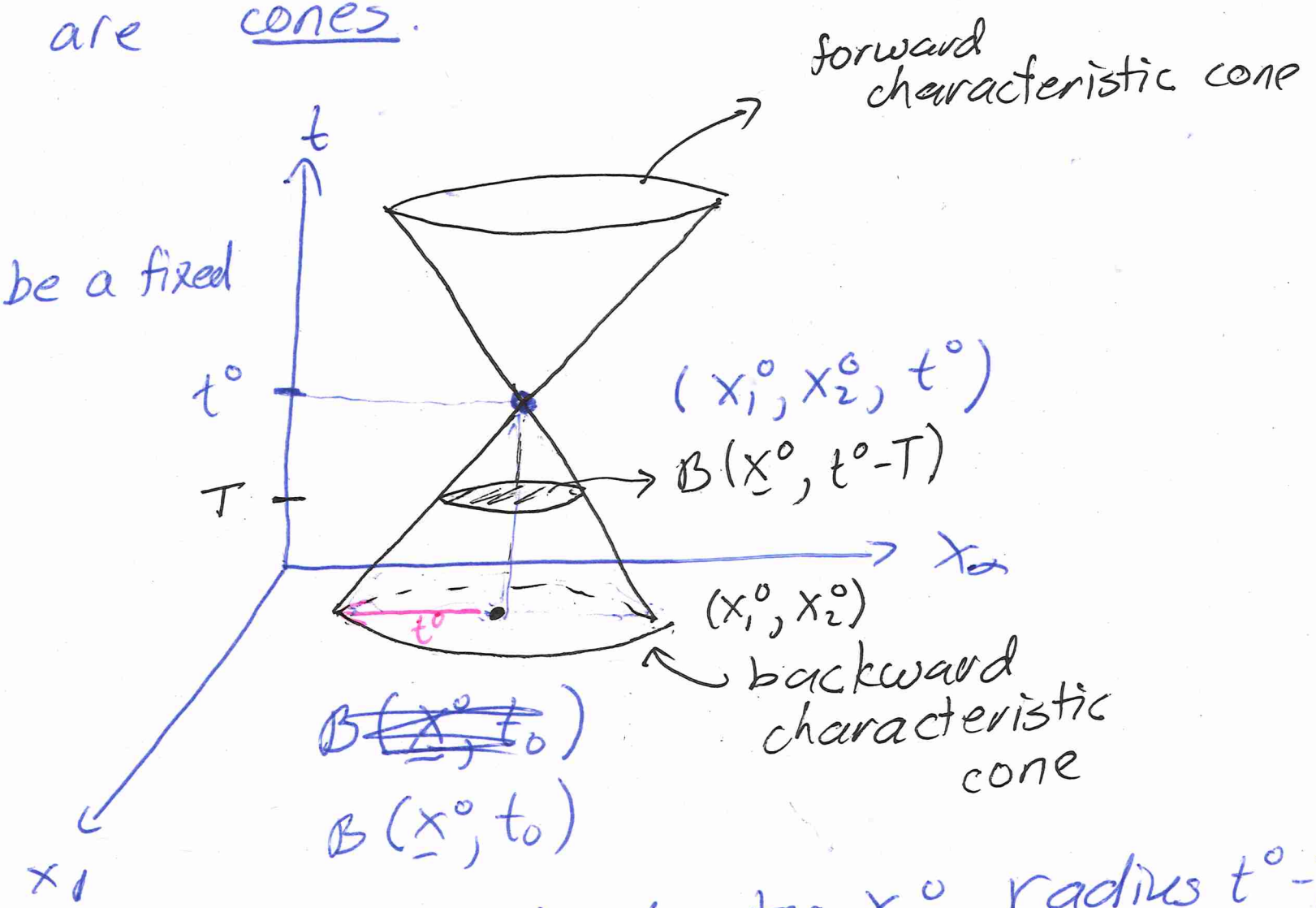
Characteristic surfaces that are $\perp$ to $\underline{\nu}$ are <u>cones</u>.

Let $T$ be a fixed time.




Define $\overline{B}(\underline{x}^0, t^0-T)$ (center $\underline{x}^0$, radius $t^0-T$) be the intersection of the plane $t=T$ and the backward cone.

Nov. 6

## § 8.3 Energy Method

<u>Thm 3.1</u> (Domain of Dependence Inequality or <u>Energy Inequality</u>)

Suppose $u \in C^2(\bar{\Omega})$ satisfies the wave eqn, where $\Omega$ is the backward cone. Then

$$\int_{\underbrace{\bar{B}(\underline{x}^0, t^0 - T)}_{\text{closure of ball at } t=T}} (u_{x_1}^2 + \ldots + u_{x_n}^2 + u_t^2)\Big|_{t=T} dx \le \int_{\underbrace{\bar{B}(\underline{x}^0, t^0)}_{\text{closure of ball at } t=0}} (\underbrace{u_{x_1}^2 + \ldots + u_{x_n}^2}_{\text{"strain energy"}} + \underbrace{u_t^2}_{\text{"kinetic energy"}})\Big|_{t=0} dx$$

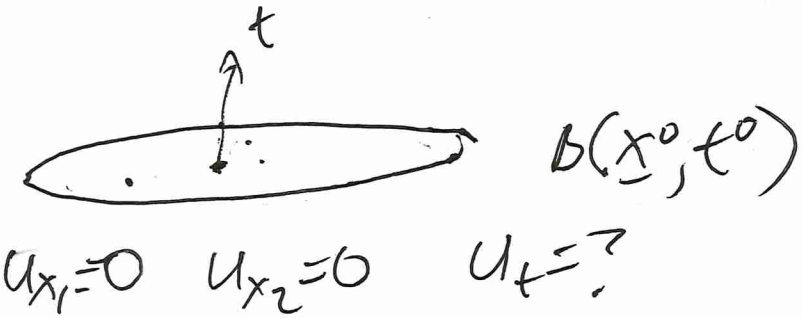
Think of integrand as being "energy", although it doesn't precisely correspond to physical energy.

So within the backward cone, energy decreases in time

## § 8.4 Consequences of Domain of Depend. Inequality

**Thm** If $u = u_t = 0$ for $\underline{x} \in B(\underline{x}^0, t^0)$ (at the base, $t=0$), then $u = 0$ in the backward cone.

**Think** If $u=0$




$u = u_t = 0$ in $B(\underline{x}^0, t)$ $\Rightarrow$ RHS ineq. is zero.

$\Rightarrow$ LHS is zero for any $T$, $0 < T < t^0$

$\int (\text{pos.})\, dx = 0$

pos. $= 0$ $\qquad u_{x_1}^2 + u_{x_2}^2 + \dots + u_{x_n}^2 + u_t^2 = 0$

$\Rightarrow u_{x_1} = u_{x_2} = \dots = u_{x_n} = u_t = 0$

$\Rightarrow u = \text{const.}$ $\qquad u = 0$ at the base,

$u \in C^2(\bar{\Omega}) \Rightarrow u = 0$ in backward cone.

<u>Cor</u> (uniqueness) The sol'n to the IVP for wave eq'n is unique.

<u>Pf</u> Assume $u_1$ & $u_2$ are sol'ns. We want to show $u_1 = u_2$.

Define $\tilde{u} = u_1 - u_2$

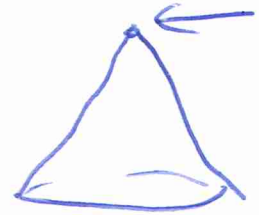
$$\Delta \tilde{u} - \tilde{u}_{tt} = 0 \qquad \underline{x} \in \mathbb{R}^n,\ t > 0$$

$$\tilde{u}(x,0) = 0 \qquad \underline{x} \in \mathbb{R}^n$$

$$\tilde{u}_t(x,0) = 0 \qquad \underline{x} \in \mathbb{R}^n$$

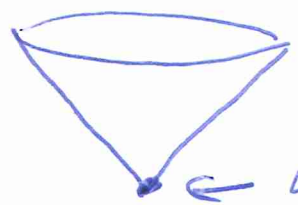
By Thm $\tilde{u} = 0$ everywhere. $\boxtimes$

The backward cone is called the <u>domain of dependence</u>



← u here only depends upon what is happening in the backward cone.

The forward cone is called the <u>range of influence</u>



← what happens here can influence the sol'n in the forward cone.

**Thm 4.2** (Conservation of Energy)

Let $u \in C^2$ be the sol'n of our IVP:

$$\Delta u - u_{tt} = 0 \qquad \underline{x} \in \mathbb{R}^n, \ t > 0$$

$$u(\underline{x}, 0) = \phi(\underline{x}) \qquad \underline{x} \in \mathbb{R}^n$$

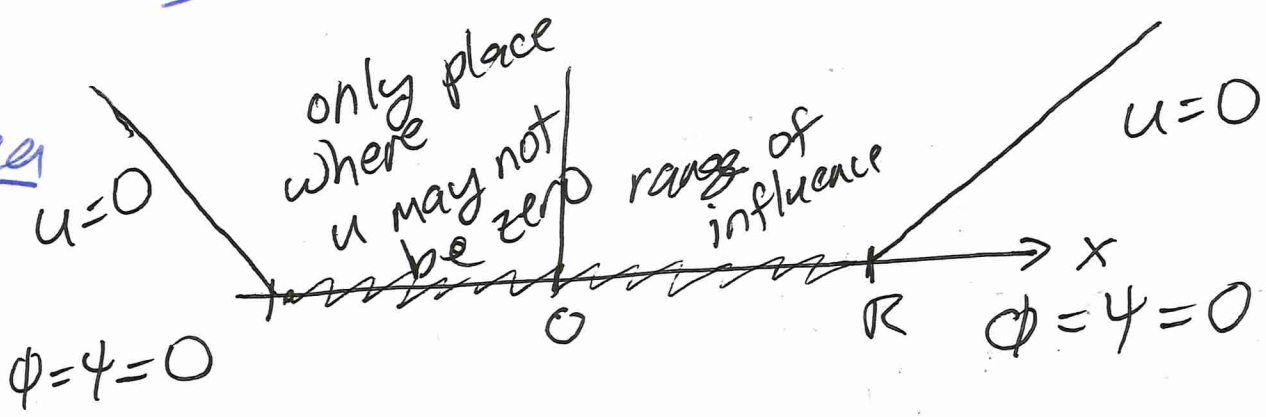
$$u_t(\underline{x}, 0) = \psi(\underline{x}) \qquad \underline{x} \in \mathbb{R}^n$$

If $\phi$ & $\psi$ are nonzero in a bounded domain (have <u>compact support</u>)

Then the energy of $u$ in $\mathbb{R}^n$ at time $T$

= the energy of $u$ in $\mathbb{R}^n$ at time $0$.

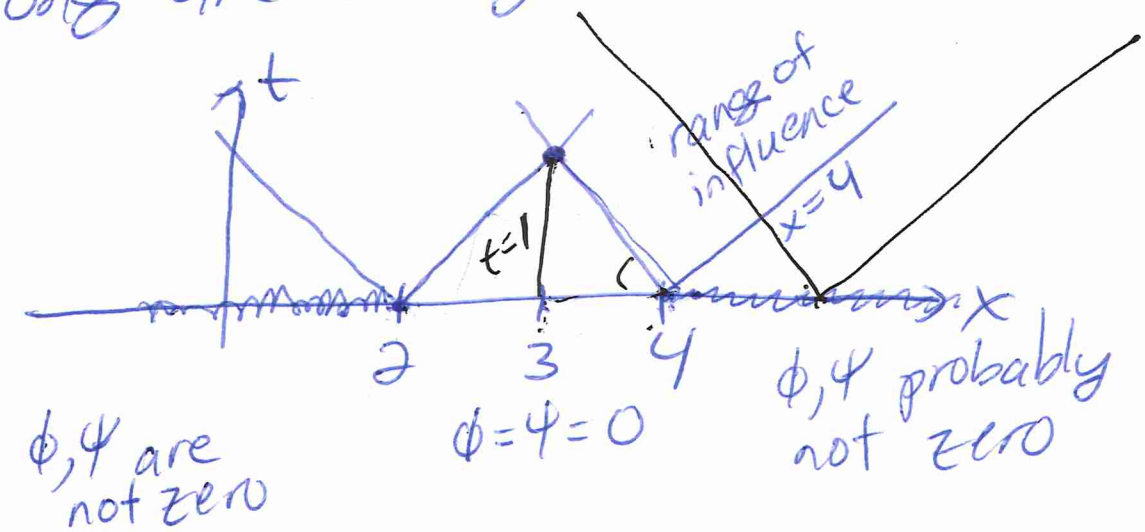
<u>Pf idea</u>



Recall problem

ex) $\phi = 0$, $\psi = 0$ $\quad 2 < x < 4$

How long are we guaranteed that $u(3,t) = 0$




$\boxed{t < 1}$

## §8.5 Existence

~~Ho~~ What is sol'n for 2 or 3D.

$n=1$ we have d'Alembert sol'n:

$$u(x,t) = \frac{1}{2}\left[\phi(x+t) + \phi(x-t)\right] + \frac{1}{2}\int_{x-t}^{x+t} \psi(x')\,dx'$$

<u>IVP</u> Find $u \in C^2(\mathbb{R}^{n+1})$ such that

$$\Delta u - u_{tt} = 0 \quad \underline{x} \in \mathbb{R}^n,\ t > 0$$

$$u(\underline{x}, 0) = \phi(\underline{x}) \quad \underline{x} \in \mathbb{R}^n.$$

$$u_t(\underline{x}, 0) = \psi(\underline{x}) \quad \underline{x} \in \mathbb{R}^n.$$

We will break this problem up into easier problems.